

JOSHUA BERGER

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Summary

Over two years as a full-stack software engineer at ICE building distributed microservices using Java, Spring Boot, PostgreSQL, and Kafka for a mortgage servicing platform that processes a majority of U.S. mortgages.

Experience

Intercontinental Exchange

August 2024 – Present

Software Engineer

Jacksonville, Florida

- Cut latency on a core create operation by 97% (10.5s to 0.4s average, 14.7s to 0.7s at p99) by replacing an application-layer duplicate check with an O(1) database index lookup, clearing the blocker that kept teams from migrating off the deprecated endpoint.
- Reduced API latency by 20+ seconds across three services by resolving compounding N+1 queries in JPA that had been hanging the UI, replacing per-entity select loops with batch loading and cutting redundant count queries by switching pagination from JPA Page to Slice; load-tested against 197,000 records with zero failures.
- Rewrote a 13,000-line backend service to replace a deprecated third-party workflow library with an in-house data layer, removing a scaling ceiling and unpatched security vulnerabilities while making new features faster to ship.
- Added asynchronous bulk actions across six operation types to an event-driven Kafka pipeline and delivered a feature a client needed for their production launch, catching a latent infinite-retry bug before it could cause an outage.
- Built an LLM-integrated tool that auto-generated configuration for 30,000+ UI fields across 1,100 legacy screens, replacing manual setup, by reverse-engineering undocumented rules from 2,000 source files; allowed 50 developers finish entering the remaining fields in a single day.
- Pinpointed the failure in a priority-1 outage within 30 seconds of walking into the room, after the team had been stuck for hours; shipped the root-cause fix across two services and later ran a team training on the debugging setup.
- Diagnosed why an integration test suite took over 1,000 seconds and rebuilt it to run in under 2 seconds, instead of patching around the timeout; raised mutation test coverage from 80% to 96% with PITest.

Intercontinental Exchange

May – August 2024

Software Engineer Intern

Jacksonville, Florida

- Selected as 1 of 3 interns (out of 50+) to receive a return offer after leading a 6-person intern team to deliver a production-ready backend service integrated with legacy mainframe systems.

Technical Skills

Languages: Java, Python

Backend & Architecture: Spring Boot, Kafka, REST APIs, OpenAPI, PostgreSQL, Liquibase, JWT-based Auth, Microservices Architecture

Tooling & Infrastructure: Docker, GitHub Actions, Git, JUnit, PITest

Projects

Automated Job Discovery & Notification System | *Python, GitHub Actions, REST APIs* February 2026 – Present

- Applied Domain-Driven Design to build a layered job aggregation system integrating concurrent REST API clients across 10+ job boards (Adzuna, Greenhouse, Lever, Workday) with deduplication and salary normalization.
- Integrated an LLM-based title filtering layer (Gemini) behind a deterministic keyword filter to reduce false-positive alerts while minimizing API cost and latency.
- Runs fully automated on GitHub Actions 3× daily, delivering results via HTML email with persistent state tracking across runs.

Education

Furman University

Bachelor of Science in Computer Science

Greenville, South Carolina

- Heatherington Scholarship recipient for outstanding academic performance (3.93 CS GPA).
- Member, Upsilon Pi Epsilon (Computing Honor Society).